

Michael Watters

Scala Engineer

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Github: <https://github.com/MichaelJohnWatters>

Summary

A Scala engineer with a background in data pipelines, custom reporting and API development. Currently building regulatory reporting for an AI-powered anti-money laundering (AML) compliance platform, submitting compliance data to government authorities in the financial crime / RegTech domain.

Experienced across the full delivery lifecycle — from writing the services in Scala and Go through Kubernetes-based deployments, GitHub Actions CI/CD, and QA automation pipelines — with a background in large-scale data processing.

Experience

Napier AI / Scala Engineer

January 2025 - Present, Belfast (Hybrid)

Scala engineer on Napier AI's financial crime compliance platform (AML / RegTech). Building regulatory reporting that takes compliance data through to submission to government authorities.

- Built regulatory reporting that takes compliance data end-to-end through to submission to government authorities.
- Extensive Kubernetes work — deploying and operating services on k8s.
- Built and maintained GitHub Actions CI/CD pipelines.
- Developed QA / test automation pipelines to safeguard release quality.

Stack: Scala, Go, Kubernetes, GitHub Actions, Kafka, PostgreSQL, Elasticsearch, SQL. QA: mabl, TestRail.

Magnite / Software Engineer II

February 2022 - December 2024, Belfast

Data Pipeline Software Engineer building large-scale batch data pipelines to onboard and normalize advertising data, for the purpose of building advertising audiences and expanding associations using third-party identity graphs.

Stack: PySpark / Spark, Databricks, Python (Luigi), Snowflake, SQL, Golang APIs, Grafana and AWS services (S3, SQS, IAM, ECS, KMS, Lambda etc).

SpotX (Acquired by Magnite) / Software Engineer I

January 2021 - February 2022, Belfast

Software Engineer, working on various custom reporting Scala applications for billing purposes and fee contracts, including a query wrapper Scala API for Apache Druid.

Stack: Functional Scala, Druid, Apache Kafka, Php, Jenkins, SQL, Openshift + Helm, Prometheus + Grafana. Developed with extensive CI/CD pipelines via Jenkins.

SpotX / Trainee Software Developer

January 2019 - January 2021, Belfast

Same as Software Engineer I.

Other

- Dart, Flutter, Flutterflow and Google Cloud Platform (Firebase etc).
- Some experience, C++ with various microcontrollers / Raspberry Pi and basic electronics, such as racing sim peripherals.
- Some experience with Java, C#, R, React, Typescript, Javascript + HTML/CSS.

Projects

Programmatic Advertising Platform / Solo Project

Personal project — <https://michaeljohnwatters.github.io>

A full programmatic advertising platform — a complete SSP / ad-exchange / DSP / ad-serving stack over OpenRTB, with a first-party data-onboarding + identity-resolution pipeline and an exactly-once billing ledger.

- Formats: display / banner (incl. interstitial, rewarded, dynamic product ads), instream video (VAST / VMAP + server-side ad insertion), native, audio (DAAST) and CTV; DOOH, retail media and in-game supported as MVPs.
- Real-time auctions across multiple strategies — single- / multi-winner, pod (with competitive separation), relevance-weighted, batch and timeslot.

Stack: Go, Kubernetes, Helm, PostgreSQL, ClickHouse, Redis, NATS JetStream, Kafka, S3/Parquet, Prometheus, Grafana. Domain: programmatic advertising, RTB / OpenRTB, SSP, DSP, ad exchange, ad serving, SSAI, VAST / VMAP, CTV, DOOH, retail media, in-game, identity resolution.

Ulster University / BSc Hons Computing Systems

September 2019 - August 2022, Belfast

Awarded 1st Degree with Honors, Bachelor of Science.

Belfast Metropolitan College / Foundation Degree, Software Engineering

September 2017 - June 2019, Belfast

Awarded 1st with distinction.

Royal Belfast Academical Institution / A-level and GCSE

September 2005 - June 2012, Belfast

A-level, AAB (Biology, Geography and History). 8 GCSEs with Math B and English C.

Education